

DevOps Practices and Principles

Mid-Term Examination Question Bank

Units 2, 3 & 4

UNIT 2 – Introduction to DevOps & DevOps Lifecycle

Very Short Answer Questions – 1/2 Marks

1. Define DevOps.
2. What does the term DevOps represent?
3. Why was DevOps introduced?
4. List any two problems of traditional software development.
5. State any two goals of DevOps.
6. What is Continuous Integration?
7. What is Continuous Delivery?
8. What is meant by continuous feedback?
9. List the phases of the DevOps lifecycle.
10. What is the role of monitoring in DevOps?
11. Who is a DevOps Engineer?
12. List any four responsibilities of a DevOps Engineer.
13. What is Infrastructure as Code (IaC)?
14. Mention any two benefits of IaC.
15. Name any two IaC tools.
16. State any two principles of DevOps.
17. What is automation in DevOps?
18. What is continuous improvement?
19. What is the main difference between Agile and DevOps?
20. Name any two tools used in the Planning phase of DevOps.

Short Answer Questions – 3/5 Marks

1. Explain the need for DevOps in modern software development.
2. Explain any five goals of DevOps.
3. Discuss the major benefits of DevOps.
4. Explain the problems associated with traditional software development.
5. Explain the DevOps lifecycle with suitable phases.
6. Explain the Planning, Coding and Building phases of the DevOps lifecycle.
7. Explain the Testing, Release and Deployment phases of the DevOps lifecycle.
8. Explain the Operations and Monitoring phases of the DevOps lifecycle.
9. Explain the role and responsibilities of a DevOps Engineer.
10. Explain the major skills required for a DevOps Engineer.
11. Explain the four major principles of DevOps.
12. Compare Agile and DevOps.
13. Explain Infrastructure as Code and its advantages.
14. Explain the role of automation in DevOps.
15. Explain how continuous feedback improves software development.

Long Answer Questions – 7/10 Marks

1. Explain the complete DevOps lifecycle with a neat diagram and describe each phase.
 2. Discuss the need, goals and benefits of DevOps in detail.
 3. Explain the role of a DevOps Engineer, including responsibilities and required technical skills.
 4. Compare traditional software development with the DevOps approach.
 5. Explain Agile and DevOps in detail and highlight their major differences.
 6. Explain Infrastructure as Code (IaC), its benefits and commonly used tools.
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UNIT 3 – Version Control Systems and Git

Very Short Answer Questions – 1/2 Marks

1. Define Version Control System.
2. Why is version control required?
3. What is a Local Version Control System?
4. What is a Centralized Version Control System?
5. What is a Distributed Version Control System?
6. Give two examples of centralized VCS.
7. Give two examples of distributed VCS.
8. What is Git?
9. Who created Git?
10. What is a Git repository?
11. Differentiate between local and remote repositories.
12. What is the staging area in Git?
13. What is the purpose of git init?
14. What is the purpose of git clone?
15. What does git status display?
16. What is the purpose of git add?
17. What is the purpose of git commit?
18. What is the purpose of git push?
19. What is the purpose of git pull?
20. What is a Git branch?
21. What is merging in Git?
22. What is a merge conflict?
23. What is .gitignore?
24. What is the purpose of README.md?
25. What is a meaningful commit message?

Short Answer Questions – 3/5 Marks

1. Explain the need for a Version Control System.
2. Compare Local, Centralized and Distributed Version Control Systems.
3. Explain the major features of Git.
4. Explain Git repository concepts.

5. Explain the Git workflow: Working Directory → Staging Area → Local Repository → Remote Repository.
6. Explain the Git lifecycle.
7. Explain the following commands with examples: git init, git status, git add.
8. Explain the following commands with examples: git commit, git push, git pull.
9. Explain git clone, git fetch and git remote.
10. Explain Git branching and its advantages.
11. Explain the process of creating and switching to a branch.
12. Explain merging in Git.
13. Differentiate between fast-forward and three-way merge.
14. What is a merge conflict? Explain how it can be resolved.
15. Explain Git repository management best practices.
16. Explain the importance of meaningful commit messages.
17. Explain the purpose and importance of .gitignore.
18. Explain the importance of protecting the main branch.
19. Explain the importance of regular git pull and frequent commits.

Long Answer / Practical Questions – 7/10 Marks

1. Explain the complete Git workflow with suitable commands and diagram.
 2. Explain Git branching and merging with a suitable example.
 3. Explain the complete process of creating a repository, adding files, committing changes and pushing them to a remote repository.
 4. Explain merge conflicts in Git with an example and describe the complete resolution process.
 5. Compare centralized and distributed version control systems and explain why Git is classified as a DVCS.
 6. Explain Git repository management and discuss best practices for collaborative software development.
 7. Write and explain the Git commands required to create a feature branch, add changes, commit them, switch to the main branch and merge the feature branch.
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UNIT 4 – Continuous Integration and Jenkins

Very Short Answer Questions – 1/2 Marks

1. Define Continuous Integration.
2. What is the main objective of CI?
3. Name any two CI tools.
4. What is Jenkins?
5. Why is Jenkins used?
6. What is Jenkins automation?
7. What is a Jenkins job?
8. What is a Freestyle project?
9. What is a Pipeline project?
10. What is a Jenkinsfile?
11. What is Pipeline as Code?

12. What is a Jenkins Controller?
13. What is a Jenkins Agent?
14. What is a Jenkins Dashboard?
15. What is build automation?
16. Name any two build automation tools.
17. What is automated testing?
18. What is continuous feedback?
19. What is a GitHub webhook?
20. What is a Jenkins trigger?
21. What is Poll SCM?
22. What are Jenkins credentials?
23. What is a pipeline stage?
24. What is the purpose of the Build stage?
25. What is the purpose of the Test stage?
26. What is the purpose of the Deploy stage?

Short Answer Questions – 3/5 Marks

1. Explain Continuous Integration and its benefits.
2. Explain the CI workflow.
3. Explain the role of Jenkins in Continuous Integration.
4. Explain Jenkins architecture.
5. Explain the Jenkins Dashboard and its major components.
6. Explain Jenkins Jobs.
7. Differentiate between Freestyle and Pipeline jobs.
8. Explain build automation in Jenkins.
9. Explain the role of Maven, Gradle or Ant in build automation.
10. What is a Jenkinsfile? Explain its purpose.
11. Explain the basic structure of a Jenkins Pipeline.
12. Explain the major stages of a Jenkins Pipeline.
13. Explain GitHub–Jenkins integration.
14. Explain the role of GitHub Webhooks in Jenkins.
15. Explain automated testing in a CI pipeline.
16. Explain continuous feedback in the CI process.
17. Explain Jenkins triggers.
18. Explain the importance of Jenkins credentials.
19. Explain the difference between manual triggering and webhook-based triggering.
20. Explain the role of build status and test reports in continuous feedback.

Long Answer / Practical Questions – 7/10 Marks

1. Explain the complete Continuous Integration workflow with a suitable diagram.
2. Explain Jenkins architecture using Controller and Agent concepts.
3. Compare Freestyle and Pipeline jobs in Jenkins with suitable examples.
4. Explain Jenkinsfile and write a basic Declarative Pipeline.
5. Explain the complete CI pipeline from GitHub commit to build, testing and feedback.
6. Explain GitHub Webhook integration with Jenkins in detail.

7. Explain build automation and automated testing in Jenkins.
8. Design a Jenkins CI pipeline containing **Checkout, Build, Test, Package and Deploy** stages.
9. Explain continuous feedback in a Jenkins CI pipeline and describe what happens when a build or test fails.
10. Describe the steps required to create a Jenkins Pipeline project and execute a Jenkinsfile.

Important Integrated Questions

These are particularly useful for a **Mid-Term Examination**, because they combine concepts from Units 1–3.

1. Explain how **DevOps, Git and Jenkins** work together in a software development lifecycle.
2. Explain the workflow:

Developer → Git → GitHub → Jenkins → Build → Test → Feedback

1. Explain how Git supports collaboration in a DevOps environment.
2. Explain how Jenkins automates the Continuous Integration process.
3. Compare the following:

Concept	Difference
Agile vs DevOps	
Git vs GitHub	
Centralized VCS vs Distributed VCS	
Freestyle vs Pipeline	
git push vs git pull	
Fast-forward vs Three-way Merge	
Manual Build vs Automated Build	

1. Design a complete DevOps workflow for a **Student Management System** using Git, GitHub and Jenkins.
2. Explain how a developer's code change can automatically trigger a Jenkins pipeline through a GitHub webhook.
3. Explain how **branching, Pull Request/code review, Jenkins CI and automated testing** can improve software quality.
4. Explain a real-world software development scenario where DevOps practices can reduce deployment failures and improve delivery speed.